1. Project Name and Description

Project Name: Library Management System (LMS)

Description and Background: The Library Management System (LMS) is designed to manage and automate the operations of a public library. This project arises from the need to transition from a primarily manual system of tracking book loans, returns, and user interactions to a streamlined, digital solution. The public library serves a diverse community of users including students, professionals, and leisure readers. The user base also includes library staff who manage the collection, assist patrons, and perform administrative tasks.

Users Involved:

- Patrons: Individuals who use the library resources, including borrowing books, accessing digital content, and attending library events.
- Librarians: Staff responsible for managing the book inventory, assisting patrons, and organizing events.
- Administrators: Library management personnel involved in strategic planning, system administration, and overseeing library operations.

2. Business Requirements

The LMS must track and manage the following information:

- Inventory: Details of books, journals, and digital media available in the library, including titles, authors, genres, publication dates, and availability status.
- Patron Information: Records of library members, including name, contact information, membership status.
- Borrowing and Returns: Transaction records for items borrowed and returned, including due dates, overdue fines, and reservation requests.
- User Access Levels: Different access levels for patrons, librarians, and administrators to secure sensitive information and functionality.

3. System Functions

- Catalog Management: Add, update, and delete items in the library’s inventory.

- Membership Management: Register new patrons, update member information, and manage membership renewals.
- Loan System: Process loans and returns, track due dates, and automatically calculate overdue fines.
- Reservation System: Allow patrons to reserve currently unavailable items and notify them upon availability.
- Reporting: Generate reports on inventory, borrowing statistics, overdue items, and event attendance for administrative use.

4. Project Plan

Project Schedule:

- Complete project proposal by 3/18
- Complete project presentation and have completed file by 4/29
- Have project paper completed by 5/6

Workload Allocation:

Divide equally between team members and complete all necessary tasks before deadlines

Project Deliverables:

- A fully functional Library Management System.
- System documentation including design documents, user manuals, and implementation guides.

5. Assumptions

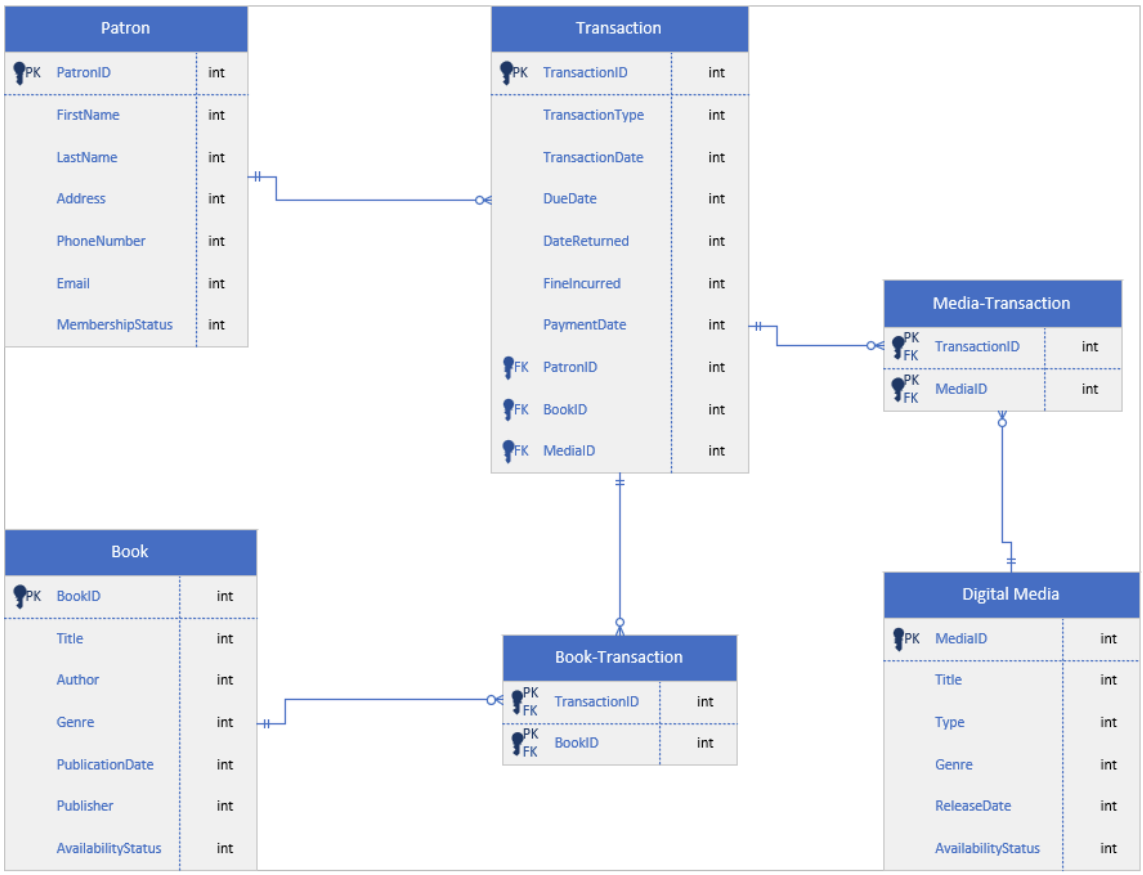
- The library operates with a stable inventory adding or removing items occasionally.
- Library patrons have internet access to utilize digital features of the LMS.
- The library staff is capable of undergoing a brief training session to learn the system.
- The LMS will be hosted on secure, reliable infrastructure with regular backups.

- Integration with existing library systems (if any) is feasible with minimal adjustments.

6. Entities and Attributes- in our

Entity- Patron
Attributes: PatronID(PK), FirstName, LastName, Address, PhoneNum,Email, MembershipStatus
Entity - Book
Attributes: BookID(PK), Title, Author, Genre, PublicationDate, Publisher, AvailabilityStatus
Entity- DigitalMedia
Attributes: MediaID(PK),Title, Type, Genre, ReleaseDate, AvailabilityStatus
Entity- Transaction
Attributes: TransactionID(PK), TransactionType, TransactionDate, DateDue, DateReturned, PaymentDate, FineIncurred, BookID (FK), PatronID (FK), MediaID (FK)

7.ER Diagram



8.Expected Forms

- Data entry forms used to input new information into the system
- Update forms can allow users to change existing data
- Search forms can allow users to find information based on a query
- Report forms can be used to alert the system of specific reports
- Transactional forms should be utilized to facilitate multi step transactions

9. Expected Reports

- Patrons who have past due items.
- Patrons who have fees due.
- Availability of items in the library.
- Expected return of items.

10. Functions we would like to add

- Search Engine